Marvin Gülhan

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Education

University of Bonn — Bonn, Germany
M.Sc. Computer Science · Focus: AI & Algorithms · Current grade: 1.1 (1.0 = best) — *Oct. 2025 – Mar. 2028*
Exchange semester: Universidad Politécnica de Madrid (UPM), Spain — *Feb. 2027 – Jun. 2027*

University of Bonn — Bonn, Germany
B.Sc. Computer Science · Focus: Algorithms · Graduated with honors · Grade: 1.2 (1.0 = best) — *Oct. 2022 – Sep. 2025*

- Thesis (Grade: 1.0): Adapted PyVRP's Hybrid Genetic Search solver to the Dynamic VRPTW and outperformed state-of-the-art ACS baselines on the majority of dynamic Solomon instances

Publications

EMNLP 2026: Reinforcement Learning Amplifies Emergent Misalignment from Harmless Rewards — *Aug. 2026*

- Showed that RL on rewards with no apparent harmful content still amplifies broad misalignment
- Implemented and evaluated mitigation techniques against RL-induced emergent misalignment

ICMCIS 2026: Multi-Agent RL for Complex and Dynamic Graph-Based Planning Problems — *May. 2026*

- Developed graph reinforcement learning with GNNs and MAPPO for dynamic multi-agent airlift planning

Experience

DHL Group — Bonn, Germany
Applied Scientist Intern — *Oct. 2026 – Dec. 2026*

- Developing routing and operational-planning optimization for same-day delivery in DHL's last-mile network
- Building package-volume forecasting models to improve capacity planning

Fraunhofer Institute FKIE — Bonn, Germany
Research Assistant — *Nov. 2023 – Sep. 2026*

- Developed online combinatorial optimization methods for dynamic routing and resource allocation under uncertainty
- Trained and evaluated GNN-based RL policies for dynamic multi-agent planning problems
- Contracted Germany-wide OpenStreetMap road graphs with 10M+ edges into solver-ready instances

Competitions

Google DeepMind: Measuring Progress Toward AGI — *Apr. 2026*

- Co-authored PDPBench, a PDPTW benchmark evaluating frontier LLMs on request insertion, route construction, and full-solution generation

ICAPS 2024: AFRL Airlift Challenge — *Apr. 2024*

- Placed 2nd in the ICAPS 2024 AFRL Airlift Challenge for dynamic cargo-delivery planning
- Designed the core routing approach for heterogeneous aircraft operating on dynamic, disrupted networks

Honors

Deutschlandstipendium (German National Scholarship) — *Oct. 2024, 2025 & 2026*

- Merit-based federal scholarship for outstanding academic achievement

Skills

Programming Languages: Python, C++
Libraries & Frameworks: PyTorch, NumPy, Pandas, Gymnasium, NetworkX
Tools: Git, Linux, Docker, Slurm
Languages: German (native), English (C1), Spanish (B1)